

Curely: Smart Health Monitoring Device

PCB Project Summary | Shawaiz Asghar | Final Year Project, ITU Lahore

Overview

Curely is a portable smart health monitoring device built around an ESP32 microcontroller. It collects vital-sign and motion data through onboard sensors and sends it over Bluetooth Low Energy to a companion Flutter mobile app. I designed and routed the main PCB of the device in KiCad, from schematic capture through board layout.

My Role

Hardware and PCB design of the main board: schematic, component selection, custom symbol creation, layout and routing, and design rule checking (DRC) in KiCad.

Hardware Design

Microcontroller	ESP32 (Wi-Fi and BLE)
Motion sensing	MPU6050 (accelerometer and gyroscope)
Pulse and SpO2	MAX30102 (optical heart-rate and oxygen sensor)
ECG	AD8232 (single-lead ECG analog front end)
Display	OLED display
Power	Battery management circuitry for a rechargeable battery
Design tool	KiCad (schematic and PCB layout)

System Context

The device pairs with a Flutter app that runs on-device AI models for health screening, which shaped the hardware requirement for reliable sensor data acquisition and stable wireless communication.

Recognition

The project secured **2nd place** at ITU's SparkUp Innovation Summit.

Skills Demonstrated

- Mixed-signal board design (analog biosignal front end alongside digital and wireless sections)
- Multi-sensor integration over shared buses on a compact, battery-powered board
- Power and battery management circuit design
- KiCad schematic capture, custom symbols, layout, routing, and DRC cleanup

Contact: shawaizasghar389@gmail.com | +92 340 6699265